VA DIVISION A • GRADES 3–4

Astronaut Lander

Official Study Guide • 2025–2026 Season

Official Materials

Scoring Formula

Maximize Flight Time

Forces on Landing

Human Body in Space

Written Assessment

📋 Event Overview — Official Rules & Materials

Rules

Build a landing device that safely protects a marshmallow astronaut (in a paper cup capsule) dropped from a two-story stairwell. Score = Assessment + Flight Time. Maximize time in the air!

MATERIALS PROVIDED

- ◆ Scissors
- ◆ 1 piece of cardboard (~10×13 cm)
- ◆ 1 — 237 ml paper cup (the capsule — MUST USE)
- ◆ 3 — index cards (8×13 cm)
- ◆ 1 meter of string
- ◆ 1 regular marshmallow (astronaut) ~2.54cm dia × 3.8cm
- ◆ 10 mini marshmallows (~0.6cm dia × 0.9cm)
- ◆ 3 rubber bands (choose from #64, #33, #19 — may mix sizes)

BUILD RULES

- ◆ 25-minute construction phase (simultaneous with written test)
- ◆ You MUST use the paper cup — it is the capsule
- ◆ May cut all materials EXCEPT marshmallows and the paper cup
- ◆ May not tape or wedge the marshmallow into the cup
- ◆ May not build any part within 10cm directly above the cup lip
- ◆ Announce estimated flight time to supervisor during build phase

SAFETY & RULES

- ◆ ANSI Z87+ impact-rated safety glasses required
- ◆ All competitors wear eye protection during ANY flight phase
- ◆ Eyeglasses are NOT safety rated
- ◆ Without proper eye protection = no-show score
- ◆ 2 participants per team number
- ◆ Coaches/spectators may NOT communicate with team during competition

🏅 Scoring Formula — Official Calculation

Scoring

Final Score (FS) = Assessment Score + Time Score. Max = 100 points. Make your lander stay in the air as long as possible — and study for the written test!

Term	Formula / Definition	Notes
Final Score (FS)	FS = Assessment Score (TAS) + Time Score (TTS)	Max 100 points
Assessment Score (TAS)	TAS = (your score ÷ best score among all teams) × 50	Max 50 pts
Time Score (TTS)	TTS = (your flight time ÷ longest flight time among all teams) × 50	Max 50 pts
Flight Time	Time from release until lander gets STUCK or comes to complete stop	Recorded to 1/100 sec
Bouncing	Time CONTINUES if lander bounces off an object	Stops when stuck/stopped
Success Condition	Marshmallow does NOT exit the cup capsule on landing = Tier 1	Exits cup = Tier 2
Tiebreaker 1	Smallest difference between your ESTIMATED and ACTUAL flight time	Estimate accuracy
Tiebreaker 2	Correct answers on selected written assessment questions	Assessment quality

Example: CAS=15, team scores 9 → TAS=(9÷15)×50=30 pts. CFS=3.56 sec, team time=1.45 sec → TTS=(1.45÷3.56)×50=20.4 pts. FS=50.4 points.

⚠ Tier Violations

Tiers

✅ TIER 1 Full Score	Team lander meets ALL building requirements. Team follows Spirit of the Problem guidelines. Marshmallow astronaut lands SAFELY (stays in the cup).
⚠ TIER 2 Score × 0.80 (−20%)	Team lander has a construction fault, OR the marshmallow astronaut exits/falls out of the cup on landing.
❌ TIER 3 Score × 0.70 (−30%)	The team did not follow the Spirit of the Problem guidelines (attempting to gain unfair advantage).

⏱️ Maximizing Flight Time — Design Strategies

Time

Flight time begins when you release the lander and continues until it stops or gets stuck. A good parachute slows descent dramatically — focus your design on maximum air resistance.

COFFEE FILTER PARACHUTE

- #6 filter is larger than #2 — provides more drag area
- Cone shape of filter traps air efficiently
- Attach string lines evenly around the filter edge for stable fall
- 4+ strings to the cup — equal lengths = no spinning/tilting
- A tilted or spinning chute is much less effective

RUBBER BAND SUSPENSION

- Rubber bands can suspend the cup below the main structure
- Elastic suspension absorbs landing impact — may help marshmallow stay in cup
- Connect rubber bands from the landing frame to the cup sides
- Test that the marshmallow does NOT slide out during slow descent
- Remember: cannot tape/wedge marshmallow into cup

USING ALL MATERIALS

- Index cards and cardboard: fold into fins to stabilize descent
- Paper straws: framework to hold the cup at a stable position
- Mini marshmallows: can serve as lightweight cushioning at base
- String: additional parachute shroud lines or structural ties
- Design must keep cup opening UPRIGHT throughout descent for marshmallow safety

🚀 Forces Experienced by Astronauts on Landing

Forces

The written assessment covers the forces and factors experienced by an astronaut when landing back on Earth after a stay in space. Study these topics carefully.

GRAVITY

Constant downward force pulling the lander toward Earth. $F = m \times g$ ($g = 9.8$ m/s^2). Gravity is what causes the fall and determines how fast the lander accelerates downward.

AIR RESISTANCE (DRAG)

Force opposing downward motion. Increases with velocity and cross-sectional area. A parachute dramatically increases drag, slowing descent. Drag = $\frac{1}{2} \times \rho \times v^2 \times Cd \times A$.

IMPACT FORCE

Sudden force on landing. Larger if impact is fast (higher velocity = more kinetic energy). Cushioning materials spread this force over more time, reducing peak force on the astronaut ($F = ma$ — more time = less force).

G-FORCE ON ASTRONAUTS

Returning astronauts experience multiple g-forces: 3g during re-entry. Muscles weaken in space — even 3g feels extreme after months in microgravity. Body must be properly oriented to distribute forces.

MICROGRAVITY EFFECTS

Astronauts in space experience near-zero gravity for months. Results: bone density loss (~1% per month), muscle atrophy, fluid shift toward head, cardiovascular changes. All make safe landing more challenging.

TERMINAL VELOCITY

Speed reached when drag force equals gravity — no more acceleration. A lander with a large parachute reaches a lower terminal velocity, meaning a gentler impact. Lower terminal velocity = safer landing.

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Human Body in Space — Study Topics

Human Body

The written assessment also covers what happens to the human body in space and during return. These are key topics that appear on the test.

BONES & MUSCLES

- Bone density loss: ~1–2% per month in microgravity
- Muscle mass decreases without the constant work of fighting gravity
- Both legs and spine are most affected — needed for walking/standing
- Astronauts exercise 2 hours/day in space to minimize losses
- Recovery time: weeks to months depending on mission length

CARDIOVASCULAR

- Heart works less in space — gravity no longer helps return blood to heart
- Heart may slightly decrease in size (less workload)
- Blood pressure changes: more fluid in upper body, less in legs
- Orthostatic intolerance on return: dizzy when standing up
- Risk: increased for heart rhythm issues during and after spaceflight

RETURNING TO EARTH

- Re-entry: astronauts experience up to 3g of force
- Lying down (prone) during landing distributes g-force across whole body
- Takes weeks to re-adapt to Earth gravity — walking feels difficult at first
- Vision can be affected: intracranial pressure changes from fluid shift
- Physical rehabilitation program required after every long-duration mission

Competition Strategy

Strategy

WRITTEN TEST (25 min simultaneous)

1. Divide roles: one student leads build, one writes test
2. Key topics: gravity, drag, impact force, g-force, microgravity effects
3. Review: what happens to bones, muscles, and heart in space?
4. Know the scoring formula: estimate accuracy is TIEBREAKER 1
5. Write your flight time estimate carefully — it matters if tied

BUILD STRATEGY

1. Must use the paper cup — design around it, not despite it
2. Coffee filter (#6) as primary parachute — attach with string from 4 points
3. Keep cup upright during descent — marshmallow must NOT fall out
4. Test with your hand: hold it by the parachute, drop from arm height — does marshmallow stay in?
5. No part of structure may extend within 10cm above the cup lip

MAXIMIZE FLIGHT TIME

1. Larger parachute area = more drag = slower descent = more time
2. Minimize weight — every gram increases fall speed
3. Attach parachute above the center of gravity for stable, non-spinning fall
4. Use rubber bands as shock absorbers at the base — may extend time if it bounces
5. Your estimate: time it during test drop — be as accurate as possible (tiebreaker)

📖 Key Vocabulary — Know These Cold

Vocab

Gravity:	Constant downward force: F = mg. Causes the fall and determines descent speed.
Air Resistance (Drag):	Force opposing motion through air. Increases with speed and area. Parachute uses this.
Terminal Velocity:	Speed where drag = gravity — no more acceleration. Larger chute = lower TV.
Impact Force:	Force at landing: reduced by increasing time of deceleration (cushioning/distance).
g-force:	Multiple of gravitational acceleration felt by the body. 1g = normal Earth, 3g = re-entry.
Microgravity:	Near-zero gravity in orbit. Not truly zero — objects are in continuous free fall.
Bone Density Loss:	Bones lose ~1–2% mass/month in space due to no load-bearing need.
Muscle Atrophy:	Muscles weaken from lack of use in microgravity. Legs most affected.
Orthostatic Intolerance:	Dizziness when standing after spaceflight — heart/circulation readapting to gravity.
Coffee Filter (#2/#6):	Provided parachute materials. #6 is larger. Cone shape traps air efficiently.
Capsule:	The 237ml paper cup — must be used, must not be cut, holds the marshmallow astronaut.
Flight Time:	Scored to 1/100 sec from release to complete stop. Includes any bouncing.